

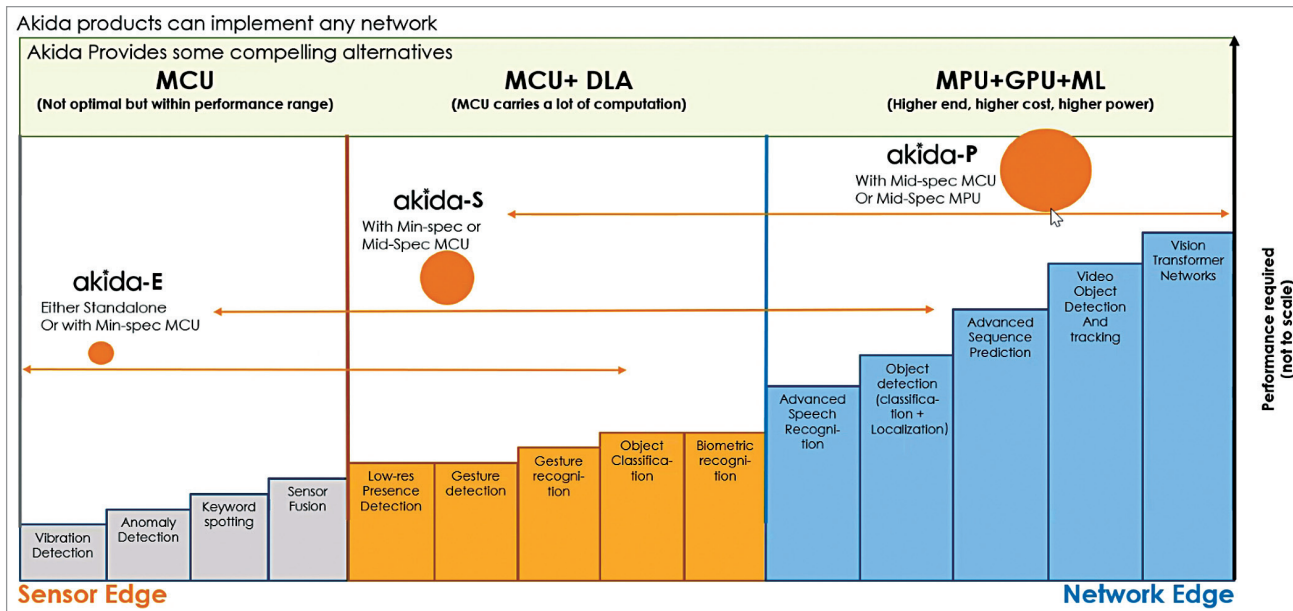


Fig. 2: Edge AI spectrum with the Akida IP Platform products. The low end done by the microcontroller unit (MCU) can be done on a small CPU, but it's not quite efficient. The mid range is addressed by the MCU with its own deep learning accelerators. The higher end workloads need a complete MCU with graphic processing unit (GPU) and ML (Credit: BrainChip)

These are a few of both the opportunities or problems that AI can solve, but also the challenges that are preventing AI from scaling to deliver those solutions. The real questions can be framed as follows:

- How do we reduce the latency and therefore improve the responsiveness of the system?
- How do we actually make these devices scale? How can we make them cost effective?
- How do you ensure privacy and security?
- How to tackle network congestion due to the large amount of data generated?

The solution

A variety of factors affect the latency of a system. For example, the hardware capacity, network latency, and the use of large amounts of data is also a problem. These devices embed AI, they also have the self-learning abilities based on the innovative time series data handling for predictive analysis. Predictive maintenance is an important factor in any manufacturing industry

as most of the industries are now turning to robotic assembly lines. In-cabin experience, for example, utilises a similar AI enabled environment, even for automated operation of autonomous vehicles (AVs). Sensing the vital signs prediction and analysis of the medical data is an important health and wellness application example. Security and surveillance as well are now turning towards AI enabled security cams for continuous surveillance.

"The more intelligent devices you make, the greater the growth of the overall intelligence."

The solution to latency lies in a distributed AI computing model, which is a strong component with the ability to embed edge AI devices, to have the performance to run necessary AI processing. More importantly, we need the ability to learn on the device to allow for secure customisation, which can be achieved by making these systems event based.

This will reduce the amount of data and eliminate sensitive data being sent to the cloud, thereby reducing network congestion, cloud

computation, and improve security. It also provides real-time responses, which makes timely actions possible. Such devices become compelling enough for people to buy as it's not only about the performance but how you make it cost-effective as well.

Neural devices are trained generally in two ways, either using the machine fed data or using spikes that mimic the functionality of spiking neural networks. This makes the system loaded.

A neural system, on the other hand, requires an architecture that can accelerate all types of neural networks, may it be convolution neural network (CNN), deep neural network (DNN), spiking neural network (SNN), or even vision transformers or sequence prediction. Therefore, using a stateless architecture in distributed systems can help reduce the load on servers and the time it takes for the system to store and retrieve data. Stateless architecture enables applications to remain responsive and scalable with a minimal number of servers, as it does not require applications to keep record of the sessions.